

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene and expansive feel.

DL for ENG

LECTURE 9.1

Laminar Flow

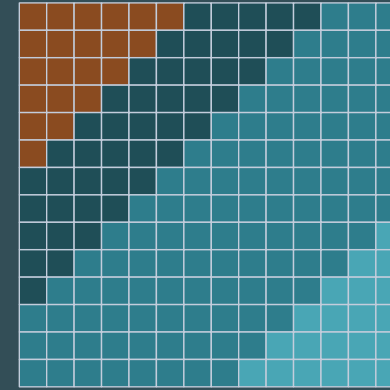
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Deep Learning for Engineering · 2026

MOMENTUM · CONTINUITY · THE FRONT



Re = 20: the front

A flow brings three things at once: a nonlinear term that carries the velocity with itself, a constraint in place of a source, and a pressure with no equation of its own. The Reynolds number says how thin the features get, and that is what decides whether a network can follow them.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Flow equations** — momentum, continuity, the Reynolds number
- 2 **Incompressibility** — a constraint, and three ways to impose it
- 3 **Solving a flow** — the outputs, the points, the checks, the ceiling

Liu carries the method and has no flow chapter, so the physics is the exercise's own: Ex_09.1's coupled Burgers equations, with an exact solution.

READ, AND WHAT EACH ANSWERS

- | | |
|----------------------|------------------------------------|
| Liu ch. 2.2 | which residuals does a flow carry? |
| Liu ch. 2.5.3 | how is a constraint built in? |
| Liu ch. 5.6 | where must the points go? |

Recap of L8.2 Transient Heat

1 - 2 · TIME

1 What does the heat equation add to steady conduction, and how does a PINN treat time?

A storage term, $\rho c_p \partial T / \partial t$, and so one initial field. The diffusivity $\alpha = k / \rho c_p$ says how fast a change spreads and sets the time constant τ . For a PINN time is one more input: the whole block of space and time is trained at once, and the solution can be read at any instant. *Liu ch. 4.1 - 4.3; Ex_08.2 sections 3 - 5*

2 What is causality, and how is it put back into training?

A solution at one instant follows from the earlier ones, but the loss treats all instants alike, so the network can fit late times first and carry an error at the start. Build the initial field into the trial solution and sample early times densely, as Ex_08.2 does; causal weights and a split window go further. *Liu ch. 4.2 - 4.4; Ex_08.2 section 5*

3 - 4 · PLATE AND INVERSE

3 How is the transient plate set up, and how is its solution checked?

The trial solution $T(x, t)$ holds the initial field and the hole wall at once; the insulated edges stay a loss term. With no exact solution, it is scored against a finite-element reference: the worst error over the plate at each instant, and the hottest point in time. *Liu ch. 4.6; Ex_08.2 sections 4 - 5*

4 How does an inverse problem differ from a forward one?

A parameter of the equation, here the diffusivity α , is trained beside the weights so that the solution fits measured data. It is optimised as $\log \alpha$, so it stays positive. Only the early part of the cooling curve constrains it: once the field has decayed, the data says nothing. *Liu ch. 4.6; Ex_08.2 section 6*

Navier–Stokes

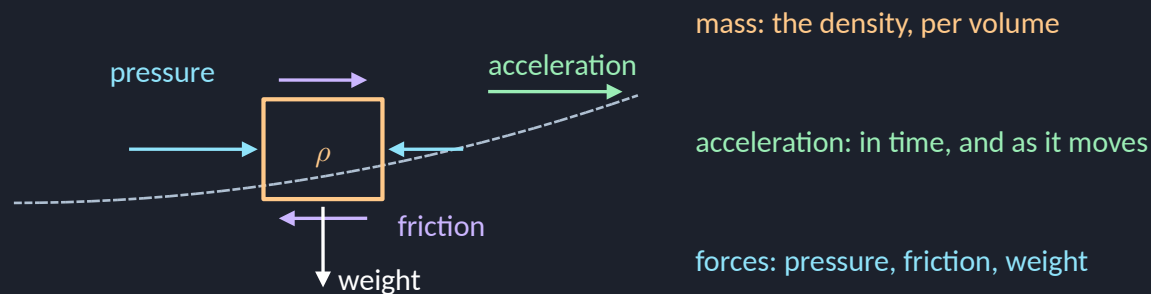
Navier–Stokes — Newton's second law for a small parcel of liquid: mass times acceleration equals the forces on it. The mass per volume is the density ρ ; the acceleration is how the parcel's velocity changes as it moves; the forces are the pressure, the friction of the neighbours, the weight.

MASS TIMES ACCELERATION EQUALS THE FORCES

$$\underbrace{\rho}_{\text{mass}} \underbrace{(\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u})}_{\text{acceleration}} = \underbrace{-\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}}_{\text{the forces}}, \quad \nabla \cdot \mathbf{u} = 0$$

ρ the mass per volume, $\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u}$ the acceleration of the parcel: in time, and because it moves on $-\nabla p$ the push of the pressure, $\mu \nabla^2 \mathbf{u}$ the friction of the neighbours ($\mu = \rho \nu$), $\rho \mathbf{g}$ the weight, $\nabla \cdot \mathbf{u} = 0$ the volume kept

A PARCEL OF LIQUID, AND NEWTON'S LAW ON IT



NEWTON'S LAW, TERM BY TERM

Newton	for a parcel of liquid	the term
mass	the density, per volume	ρ
acceleration	the velocity changes in time	$\mathbf{u}_t$
	and because it moves on	$(\mathbf{u} \cdot \nabla)\mathbf{u}$
force	the pressure, from the sides	$-\nabla p$
	the neighbours drag on it	$\mu \nabla^2 \mathbf{u}$
	its weight	$\rho \mathbf{g}$

Why $(\mathbf{u} \cdot \nabla)\mathbf{u}$ is an acceleration: in a time dt the parcel moves from x to $x + \mathbf{u} \cdot dt$, where the velocity is different, so its velocity changes by $\mathbf{u}_t \cdot dt + (\mathbf{u} \cdot dt \cdot \nabla)\mathbf{u}$. In a steady flow through a narrowing pipe $\mathbf{u}_t = 0$, yet every parcel speeds up.

Inertia and Darcy

Inertia term — $(\mathbf{u} \cdot \nabla) \mathbf{u}$, the one term where the velocity multiplies itself: the fluid carries its own momentum. Double $\mathbf{u}$ and every other term doubles, this one quadruples. Solutions can no longer be added, large eddies feed smaller ones, and the flow can turn turbulent.

FRAME IT, NEGLECT IT, AVERAGE IT

$$\rho \mathbf{u}_t + \boxed{\rho (\mathbf{u} \cdot \nabla) \mathbf{u}} = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

keep it: nonlinear, turbulence (L9.2)

$$\rho \mathbf{u}_t = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

neglect it: Stokes, linear, slow flow

$$\bar{u} = -\frac{K}{\mu} \frac{dp}{dx}, \quad K = \frac{h^2}{12}$$

average across the channel: Darcy
the law this lecture rests on

$(\mathbf{u} \cdot \nabla) \mathbf{u}$ the inertia, the one nonlinear term; $\bar{u}$ the mean speed across the channel, dp/dx the pressure drop per length K the permeability, $h^2/12$ for a channel of height h , μ the viscosity, $\rho \mathbf{g}$ the weight

WHY IT IS NONLINEAR: DOUBLE THE VELOCITY

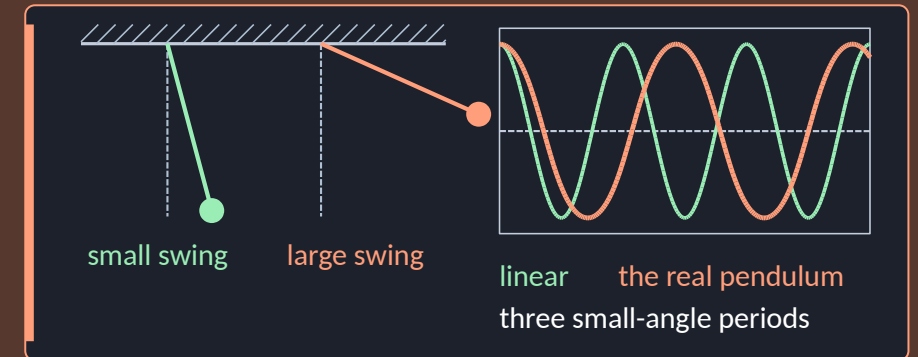
$\rho \mathbf{u}_t, \mu \nabla^2 \mathbf{u}, \nabla p$ double: linear

$\rho (\mathbf{u} \cdot \nabla) \mathbf{u}$ quadruples: $u \cdot u_x = (u^2/2)_x$

THE SAME STEP ON A PENDULUM

$$\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0 \Rightarrow \ddot{\theta} + \frac{g}{\ell} \theta = 0$$

θ the angle, ℓ the length, g gravity; $\sin \theta \rightarrow \theta$ is the same step as dropping the inertia



Small swing, $\sin \theta \approx \theta$: linear, one period at any amplitude, like Stokes. Large swing: the period grows with the amplitude, and driven it goes chaotic, like Navier–Stokes; the amplitude plays Re.

The Reynolds Number

Reynolds number — the ratio of convection to viscosity, $Re = UL/\nu$. Scaled by the device size and the inlet speed, it is the only parameter left in the equations, and it sets how thin the features of the solution are: in Ex_09.1 the front's width is $8/Re$.

ONE NUMBER, AND IT SETS A LENGTH

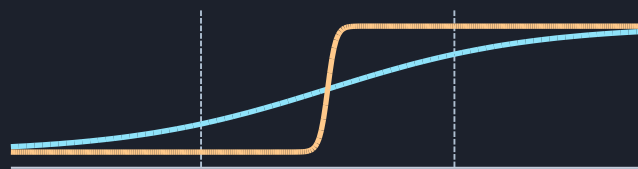
$$Re = \frac{UL}{\nu}$$

$$\delta = 8\nu = \frac{8}{Re}$$

$$\mathbf{u}_t^* + (\mathbf{u}^* \cdot \nabla) \mathbf{u}^* = -\nabla p^* + \frac{1}{Re} \nabla^2 \mathbf{u}^*$$

U the inlet speed, L the device size, δ the width of the front; starred: scaled by U and L

THE FRONT OF EX_09.1, AT TWO REYNOLDS NUMBERS



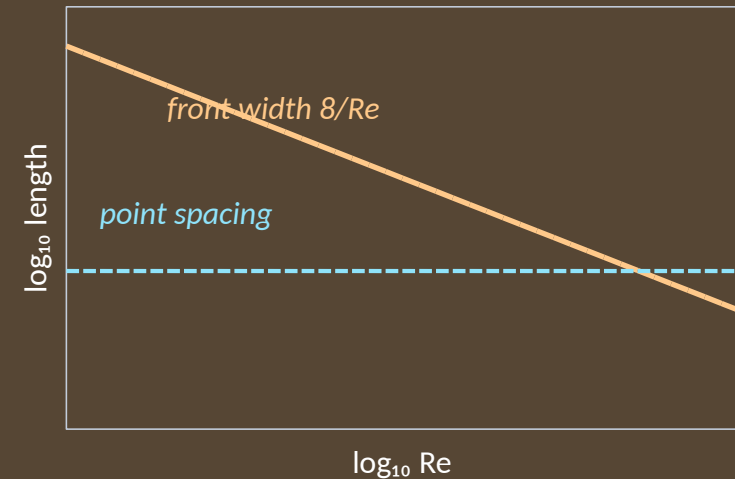
the square

$Re = 20$: the front is 0.40 wide

$Re = 500$: it is 0.016 wide

the same jump, a thinner front

FRONT WIDTH AGAINST Re



Cyan is the spacing of 4 000 points on a unit square. The amber line crosses it at $Re = 500$, where the front is as thin as the sample: Ex_09.1 section 6.

Coupled Burgers

Coupled Burgers equations — Navier–Stokes divided by ρ , without pressure, weight and constraint. The convective term $u \cdot u_x + v \cdot u_y$ is the inertia $(\mathbf{u} \cdot \nabla) \mathbf{u}$ written out: it carries the front and steepens it, ν smears it, their balance sets its width. Nonlinear, with an exact solution.

THE STEPPING STONE OF EX_09.1

$$u_t + u u_x + v u_y = \nu (u_{xx} + u_{yy})$$

$$v_t + u v_x + v v_y = \nu (v_{xx} + v_{yy})$$

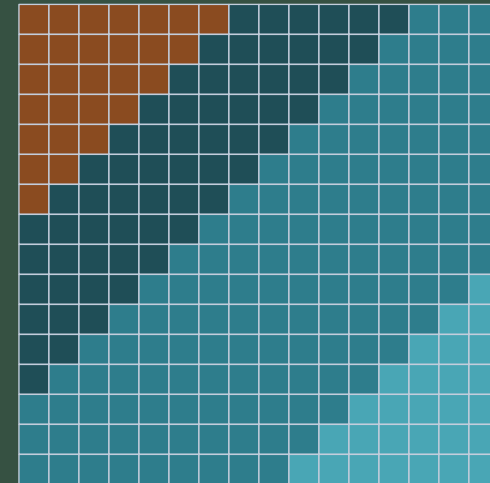
$$u = \frac{3}{4} - \frac{1}{4} \sigma\left(\frac{y-x-t/4}{8\nu}\right), \quad u + v = \frac{3}{2}$$

σ the logistic function, ν the viscosity; the front lies along $y = x + t/4$

COUPLED BURGERS IN PYTORCH

```
u, v = split(net(xyt))          # two outputs, one network
F_u = grad(u,t) + u*grad(u,x) + v*grad(u,y) - nu*lap(u)  # v enters here
F_v = grad(v,t) + u*grad(v,x) + v*grad(v,y) - nu*lap(v)  # and u here
```

u OVER THE SQUARE, AT t = 0



$Re = 20$: the front is 0.4 wide

convection

$u \cdot u_x + v \cdot u_y$, the inertia term

the front

along $y = x + t/4$, width $8/Re$

the identity

$u + v = 3/2$, everywhere

Flow Boundary Conditions

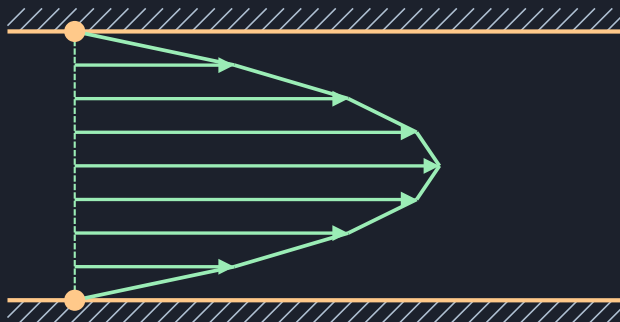
Flow boundary conditions — a wall carries no-slip, $\mathbf{u} = 0$, because a viscous fluid does not slide along it. An inlet prescribes the velocity profile, an outlet prescribes the pressure and lets the velocity leave, and the pressure needs one reference value somewhere.

FOUR KINDS OF EDGE

$\mathbf{u} = 0$ on the wall, $u = U(y)$ at the inlet

$\mathbf{u}$ the velocity, $U(y)$ the inlet profile

BETWEEN TWO WALLS



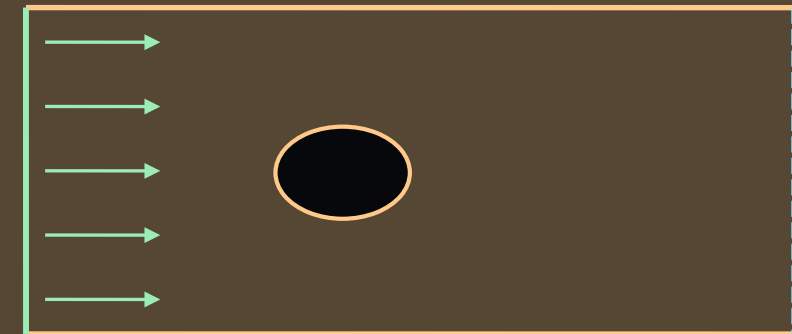
on a wall the fluid does not move

so the speed rises from zero

the outlet sets the level of p

THE CHANNEL OF EX_09.2

walls and obstacle: no-slip



inlet

outlet

no-slip

both components zero on a wall

inlet

a profile, not a constant

outlet

pressure fixed, velocity free

Imposing Incompressibility

Incompressibility — the constraint $\nabla \cdot \mathbf{u} = 0$, which every instant of the flow must satisfy. It can be asked for as a loss term, built into the network through a stream function, or projected onto after each step. Only the second holds it exactly and for free.

THREE WAYS, AND WHAT EACH COSTS

- **a penalty** — one more term, one more weight
- **a stream function** — exact, and one output fewer
- **a projection** — exact, and a solve at every step

TALK • 2 MIN How would you explain what incompressibility asks of a velocity field, in your own words? Then, in the table on the right, which of the three would you choose for the channel of Ex_09.2, and what does it cost you?

THE THREE ROUTES

route	holds	costs
penalty term	roughly	a weight to tune
stream function	exactly	a derivative more
projection	exactly	a linear solve

The continuity residual has no time derivative and is the smallest term in the sum, so as a penalty it is the term most easily outvoted. That is the argument for building it in.

Stream Function

Stream function — a single scalar ψ whose derivatives are the velocity: $u = \psi_y$ and $v = -\psi_x$. Continuity then holds identically, because the two cross derivatives cancel, so the network outputs ψ instead of u and v and the loss has no continuity term at all.

ONE OUTPUT INSTEAD OF TWO

$$u = \psi_y, \quad v = -\psi_x$$

$$\nabla \cdot \mathbf{u} = \psi_{yx} - \psi_{xy} \equiv 0$$

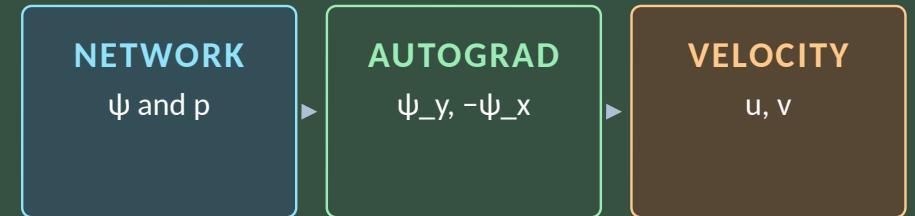
ψ the stream function, u, v the velocity components it gives

- **the price** — one derivative more in every residual
- **the gain** — no weight, and no residual to balance

THE STREAM FUNCTION IN PYTORCH

```
psi, p = split(net(xy))           # one scalar, plus the pressure
u, v = grad(psi, y), -grad(psi, x) # continuity holds identically
```

FROM ONE SCALAR TO A FLOW



outputs ψ and p , not u, v and p

continuity gone from the loss

derivatives third order in the residual

The momentum residual differentiates u twice and u is already a derivative of ψ , so the graph is one order deeper. That is the whole price.

Pressure Reference

Pressure reference — only the gradient of the pressure appears in the momentum equation, so adding a constant to p leaves the flow unchanged. A network will drift along that freedom unless one value is fixed: a reference point, or the pressure at the outlet.

THE SAME FLOW, A DIFFERENT LEVEL

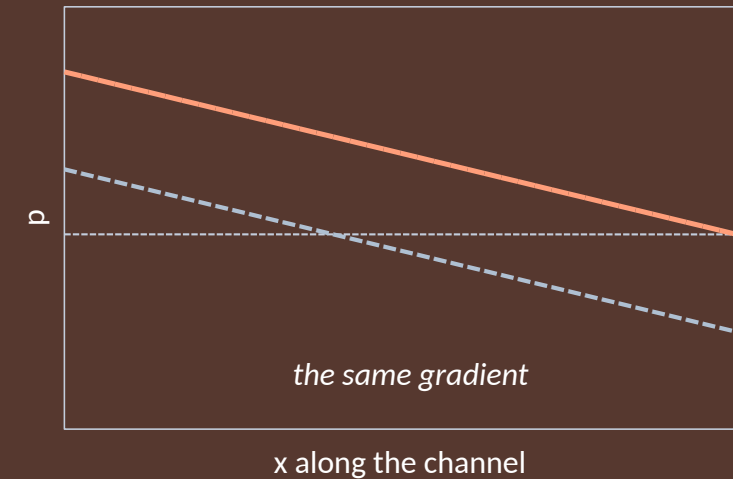
$p \rightarrow p + c \Rightarrow$ the same velocity field

p the pressure, c any constant: only ∇p enters the equations

- **the symptom** — a pressure that wanders during training
- **the fix** — one value pinned, usually at the outlet



TWO PRESSURES, ONE FLOW



Both drive exactly the same flow. Only the gradient is physical, so quote a pressure difference, never a pressure.

Shared Network

Shared network — one network with several outputs, rather than one network per field. The hidden layers are shared, so the coupling between the components is represented inside the network, and the parameter count is spent once instead of twice.

SHARED LAYERS, SEVERAL HEADS

$$(\psi, p) = \mathcal{N}_{\hat{w}}(x, y, t)$$

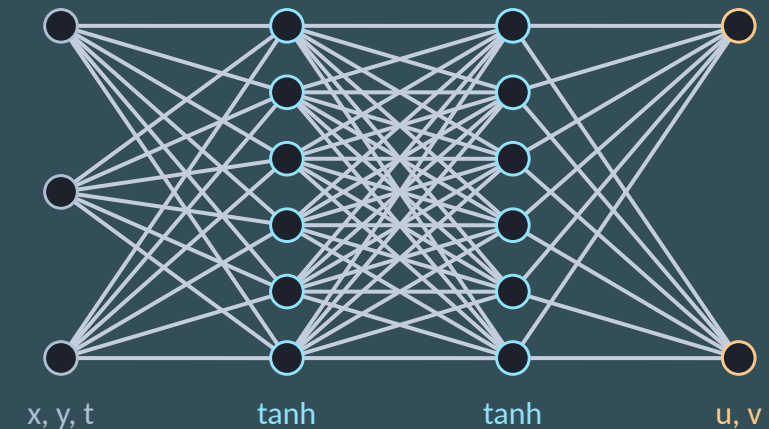
ψ the stream function, p the pressure, $\mathcal{N}_{\hat{w}}$ one network for both

- **shared** — the coupling is learned once
- **separate** — the same budget spent twice

THE NETWORK IN PYTORCH

```
net = MLP(n_in=3, n_hidden=32, n_layers=4, n_out=2) # (x, y, t) in, two out
u, v = net(xyt).split(1, dim=1)                    # the two components
loss = F_u.pow(2).mean() + F_v.pow(2).mean() + L_bc # one loss, two residuals
```

ONE TRUNK, TWO OUTPUTS



Ex_09.1 trains this network: four layers of 32, two outputs, and the coupled term of the residuals is why the layers are shared.

Where the Points Go

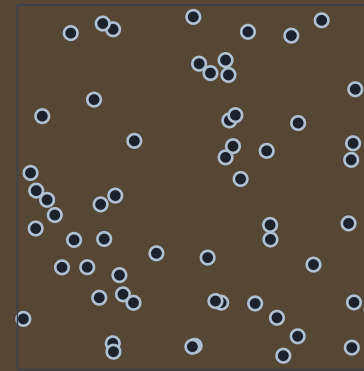
Sampling a flow — a flow puts its difficulty in thin places: a front, a boundary layer, a wake. A uniform sample spends almost every point where the solution is flat, so points are graded towards the walls and refined where the residual is largest.

THE POINTS BELONG WHERE THE FIELD BENDS

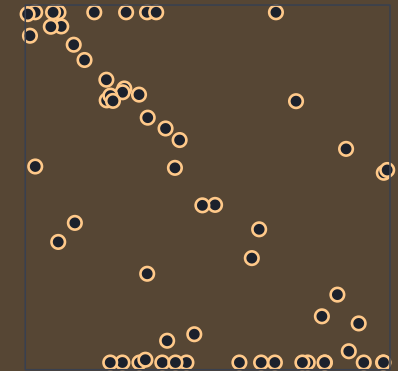
- **uniform** — the error plateaus, whatever you add
- **near the wall** — graded, because the layer is thin
- **at the front** — refined where the residual is large

TALK • 2 MIN How would you decide where a solver should spend its effort, in your own words? Then, in the two squares on the right, which sample would you trust at $Re = 500$, and what would you measure to be sure?

THE SAME BUDGET, TWO WAYS



uniform



refined at the front

Sixty points each. On the left almost all of them sit where u is flat and already right; on the right they follow the diagonal, where it is not.

Checking a Flow

Checking a flow — a flow gives several checks that need no exact solution: the continuity residual, an identity the solution must obey, the mass through any cross-section, and a profile against a reference. A low loss with a smooth wrong field passes none of them.

FOUR CHECKS, NONE OF THEM THE LOSS

$$\mathcal{L} = \mathcal{L}_u + \mathcal{L}_v + w_c \mathcal{L}_{cont} + \mathcal{L}_{BC} + \mathcal{L}_{IC}$$

$\mathcal{L}_u, \mathcal{L}_v$ the momentum residuals, $\mathcal{L}_{cont}$ continuity, w_c its weight

- **the loss** — says the residual is small, and nothing else
- **the checks** — say the field is a flow

THE CHECKS IN PYTORCH

```
div = (grad(u,x) + grad(v,y)).abs().mean()    # continuity, after training
ident = (u + v - 1.5).abs().max()             # Ex_09.1's identity
mass = trapz(u(x0, ys), ys)                   # mass through a section
err = (pred - exact).norm() / exact.norm()    # only where there is an exact
```

WHAT EACH CHECK CATCHES

check	catches
continuity	a field that is not a flow
the identity	a front in the wrong place
mass balance	a leak through a wall
a profile	the wrong shape, at the right size

A smooth, plausible, wrong field is more dangerous than a crash, because nothing in the training log looks unusual.

The Reynolds Ceiling

Reynolds ceiling — the Reynolds number at which a given sample stops resolving the solution. Below it the error falls with training; above it the front is thinner than the spacing between points, the loss still falls, and the field is smooth and wrong.

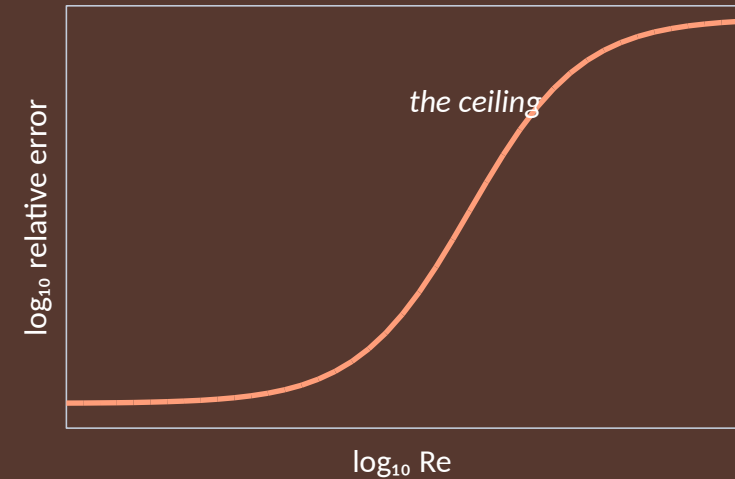
THE LOSS KEEPS FALLING, THE FIELD DOES NOT

- **below the ceiling** — more points, less error
- **above it** — more points, the same error
- **the fix** — refine at the front, or drop Re

THE SWEEP IN PYTORCH

```
for Re in [20, 500]:  
    net = train(nu=1/Re, n_points=4000)  # Ex_09.1 section 6  
    log(Re, 8/Re, rel_error(net))        # everything else fixed  
                                         # width, and what it cost
```

SCHEMATIC • ERROR AGAINST Re



Schematic. In Ex_09.1 the grid meets it: 6 nodes a side at $Re = 20$, 321 at $Re = 500$. The network does not: that front is one tanh.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
adaptive refinement	points where the residual is large	Liu ch. 5.6	thin fronts, high Re
time windows	the transient in pieces	Liu ch. 4.2 – 4.3	long unsteady runs
inverse flow	viscosity or pressure from data	Liu ch. 2.2	measured velocity fields
<u>porous media</u>	Darcy flow: no convection	Liu ch. 2.4	groundwater, CO ₂ storage
turbulent mean flow	an effective viscosity	Pope ch. 4, 10	any real Reynolds number

WHY THEY ARE LEFT OUT

Ex_09.1 runs at two Reynolds numbers, with an exact solution to score against, so none of them is needed to do it. The last row is the next lecture, and it is the row that decides whether any of this reaches a real flow.

Exercise 9.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — a front in a flow: data, physics, scaled
- **4** — the exact solution, and finite differences
- **5** — a PINN with two outputs; the two compared
- **6** — the Reynolds number raised from 20 to 500
- **7 – 9** — the report; mini projects MP9.1A and MP9.1B

THE QUESTION EACH PART ANSWERS

- 1 – 3** how thin is the front?
- 4** which grid is fine enough?
- 5** how does a PINN compare?
- 6** what does $Re = 500$ cost?

REPORT QUESTION, TO CONCLUDE

At $Re = 500$ the grid needed 321 nodes a side, and the network's error did not grow. Which method would you hand to a colleague at each Reynolds number, and how far does the network's result carry to other flows?

Key Takeaways

NAVIER-STOKES

what it is	momentum, plus a constraint
function	the velocity, and a pressure that enforces
train, tune	scale first; Re is the only number left
vs heat	nonlinear, coupled, constrained
used for	channels, wakes, anything that moves

SOLVING A FLOW

what it is	one network, several outputs
function	ψ for continuity, p for the level
train, tune	points at the front and the wall
vs uniform points	the error stops falling
used for	up to the ceiling $8/Re$ sets

Next — L9.2: the same equations at a Reynolds number no sample can resolve.

Questions

1 - 2 • NAVIER-STOKES

1 What are the incompressible flow equations, and what does the pressure do?

Momentum, which says how the velocity changes, and continuity, which says the fluid is not compressed. The convective term is nonlinear and couples the components. The pressure has no equation of its own: it is whatever makes continuity hold, and is known only up to a constant. *Liu ch. 2.2; Ex_09.1 sections 3, 5*

2 What does the Reynolds number tell you, and how is incompressibility imposed?

It is the ratio of convection to viscosity, the only parameter left after scaling, and it sets how thin the features of the flow are. Continuity can be a loss term, which is easily outvoted, or be built in by a stream function, which is exact at the price of one more derivative. *Liu ch. 2.5.3; Ex_09.1 section 6; Ex_09.2 section 5*

3 - 4 • SOLVING A FLOW

3 How is a flow network built, and where do its points go?

One network with several outputs, so that the coupling is represented in the shared layers. A flow keeps its difficulty in thin places, a front or a boundary layer, so the points are graded towards them; uniform points stop helping once the front is thinner than their spacing. *Liu ch. 2.1, 5.6; Ex_09.1 sections 5 - 6*

4 How do you check a flow that has no exact solution?

By the continuity residual, the mass through any cross-section and a profile against a reference. Above the Reynolds number its points can resolve, the loss still falls and the field is smooth and wrong. That is worse than a crash, because nothing warns you. *Ex_09.1 sections 5 - 6*

NEXT

L9.2 — Turbulent Flow

What happens above the ceiling: averaged equations, a closure that has to be modelled, and the one use where the method is clearly the right tool.

BEFORE THE NEXT SESSION

read	Liu ch. 2.5.3, 5.6
run	Ex_09.1, sections 1 to 7
bring	your errors at $Re = 20$ and 500
think about	the thinnest feature in your problem

*A laminar flow is solved, and its limit is measured.
Next, the flows that are above that limit by
construction.*